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BuildOcc: A Large Language Model Occupant Agent Platform for Building Energy Research --Manuscript Draft--

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Abstract:	Occupants are a primary source of uncertainty in building energy consumption and management, yet existing occupant behavior models cannot capture adaptive and reasoning responses considering the occupant's personal history, current context, and the type of energy signal being delivered. This study presents BuildOcc, an open-source Python platform that grounds large language model agents in the American Time Use Survey (ATUS), a nationally representative diary dataset covering 16,684 respondents. Through BuildOcc, each simulated occupant agent can be instantiated with a demo-graphic persona drawn from ATUS population statistics, a memory stream that accumulates and reflects on timestep-level observations, and an activity scheduler that samples empirically from ATUS time-at-activity distributions. The platform exposes a three-layer interface—Python library, REST API, and Model Context Protocol server—so that any building energy tool (EnergyPlus, Home Assistant) can integrate behavioral intelligence without bespoke coupling code. A plugin registry lets the community add new occupant strata, custom schedulers, and alternative memory backends as separate installable packages. Two validation tiers confirm that ATUS-grounded sampling produces empirically calibrated activity distributions and that demographic priors propagate into coherent, persona-consistent agent reasoning across timesteps. BuildOcc is expected to advance how the human behavioral layer is modeled in building energy research, giving the community a shared foundation for more realistic and demographically representative studies of occupant-driven energy use. BuildOcc is openly released at https://doi.org/10.5281/zenodo.21192895 under the Apache License 2.0 and installable via <code>pip install buildocc</code>.

BuildOcc: A Large Language Model Occupant Agent Platform for Building Energy Research

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Abstract

Occupants are a primary source of uncertainty in building energy consumption and management, yet existing occupant behavior models cannot capture adaptive and reasoning responses considering the occupant's personal history, current context, and the type of energy signal being delivered. This study presents BUILD OCC, an open-source Python platform that grounds large language model agents in the American Time Use Survey (ATUS), a nationally representative diary dataset covering 16,684 respondents. Through BuildOcc, each simulated occupant agent can be instantiated with a demographic persona drawn from ATUS population statistics, a memory stream that accumulates and reflects on timestep-level observations, and an activity scheduler that samples empirically from ATUS time-at-activity distributions. The platform exposes a three-layer interface—Python library, REST API, and Model Context Protocol server—so that any building energy tool (EnergyPlus, Home Assistant) can integrate behavioral intelligence without bespoke coupling code. A plugin registry lets the community add new occupant strata, custom schedulers, and alternative memory backends as separate installable packages. Two validation tiers confirm that ATUS-grounded sampling produces empirically calibrated activity distributions and that demographic priors propagate into coherent, persona-consistent agent reasoning across timesteps. BUILD OCC is expected to advance how the human behavioral layer is modeled in building energy research, giving the community a shared foundation for more realistic and demographically representative studies of occupant-driven energy use. BUILD OCC is openly released at <https://doi.org/10.5281/zenodo.21192895> under the Apache License 2.0 and installable via `pip install buildocc`.

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Keywords: Artificial Intelligence Agent, Occupant Behavior Modeling, Building Energy Use and Management, American Time Use Survey,

Required Metadata

Current code version

Table 1: Code metadata (mandatory)

Nr.	Code metadata description	Please fill in this column
C1	Current code version	v1.0.0
C2	Permanent link to code/repository	https://github.com/humanbuildingsynergy/BuildOcc
C3	Permanent link to reproducible capsule	https://doi.org/10.5281/zenodo.21192895
C4	Legal code license	Apache License 2.0
C5	Code versioning system	Git
C6	Software code languages, tools, and services used	Python 3.11+
C7	Compilation requirements, operating environments & dependencies	<code>pip install buildocc</code> ; requires an LLM API key (Anthropic, OpenAI, Google), or local Ollama
C8	Developer documentation	https://github.com/humanbuildingsynergy/BuildOcc/blob/main/README.md
C9	Support email for questions	wooyoungjung@arizona.edu

1. Motivation and Significance

Humans are complex systems that respond to environmental, social, and economic signals in ways that are difficult to predict, introducing substantial uncertainty in building energy use and management [1, 2]. This long-standing challenge has motivated the development of occupant behavior models that can capture the stochastic and adaptive nature of human decision-making [3]. However, most existing models rely on fixed schedules (e.g., ASHRAE 90.1 reference schedules [4]) or stochastic processes that do not comprehensively

capture and reason about the occupant’s personal history, current context, or the type of energy signal being delivered [3, 2]. Consequently, these models cannot replicate the nuanced and adaptive responses that real occupants exhibit in response to various conditions.

15 The emergence of large language models (LLMs) and their architecture for reasoning and memory has opened new opportunities for artificial intelligence (AI) agents to simulate human-like decision-making processes. These agents are capable of contextual reasoning, memory retention, and adaptive behavior, making them suitable for modeling complex systems, including occupants in buildings. Recent studies demonstrated the potential of LLM-integrated agents to simulate occupant behavior in building energy contexts, showing that they can produce more realistic and adaptive responses to various energy signals compared to traditional models [3, 5, 6, 7, 8]. Specifically, Nicholas et al. [9] proposed agents that adopt the generative agent memory architecture of Park et al. [10]—combining a persona-based core memory, a timestep-level memory stream, and periodic reflection—and demonstrate that this representation produces qualitatively different energy demand patterns than fixed-schedule or rule-based occupant models. Other recent approaches include agents conditioned on IEQ sensor readings for comfort and health feedback [11], personality-driven agents for negotiating shared thermostat setpoints [12], and a Thought-Action-Reflection-Journal framework for modeling energy behavior under dynamic pricing [13]. However, these agents are instantiated from hand-crafted profiles or small convenience samples [14]. In addition, none provides quantitative validation against empirical activity benchmarks, and none releases an open-source platform for integration with building energy tools—leaving the community without a general-purpose, survey-grounded foundation for reproducible occupant behavior research.

This study aims to fill this gap by introducing BUILD OCC, an open-source platform for LLM-based occupant behavior simulation and building energy management. BUILD OCC grounds agent personas and activity schedules in the American Time Use Survey (ATUS) [15], a nationally representative dataset covering tens of thousands of diary respondents across income, employment, and household-type strata. Each agent maintains a memory stream that accumulates timestep-level observations and periodically synthesizes higher-order behavioral insights through reflection—enabling coherent, adaptive decision-making across a simulation day rather than independent per-timestep responses. Users can drive agents through building simulation timesteps using any supported LLM provider and receive structured behavioral decisions—thermostat adjustments, device control actions, and demand response reactions—that feed directly into EnergyPlus [16] co-simulation, Home Assistant automations, or custom building management

systems through a three-layer interface (Python library, REST API, and MCP server). Abstract base classes and a plugin registry allow the community to extend the platform with new occupant strata, behavioral data sources, and memory architectures. Consequently, BUILD OCC is positioned to advance how the human behavioral layer is modeled in building energy research, giving the community a shared foundation for more realistic and demographically representative studies of occupant-driven energy use.

2. Software Description

Following an overview of the platform architecture, the platform is described in five parts: the ATUS behavioral grounding that drives activity scheduling and persona sampling; the inputs required at simulation initialization; the per-timestep LLM reasoning pipeline; the demand response signal interface; and the extensibility mechanisms that allow the platform to be adapted to new research contexts.

2.1. Architecture Overview

BUILD OCC is organized around two levels of architecture: an internal *agent component architecture* that defines how occupant agents perceive, reason, and act within a simulation timestep, and a *platform interface* that exposes this functionality to external building energy tools (Figure 1).

Agent Component Architecture. The BUILD OCC occupant agent is composed of six interacting components:

- **Persona** — a structured demographic profile assigned to one of four strata (O1–O4; Table 2), with appliance ownership, income bracket, and thermal comfort tolerance sampled from RECS [17] probability distributions.
- **ActivityScheduler** — samples the occupant’s activity at each 15-minute timestep from stratum- and hour-specific probability distributions derived from ATUS 2022–2023 diary records (16,684 respondents), with weekday/weekend stratification (Appendix A).
- **MemoryStream** — an accumulating log of observations, actions, and reflections ranked by recency and LLM-assigned importance following Park et al. [10]; top- k entries are injected into each timestep prompt and periodically synthesized into higher-order behavioral insights through a reflection call (Appendix B).

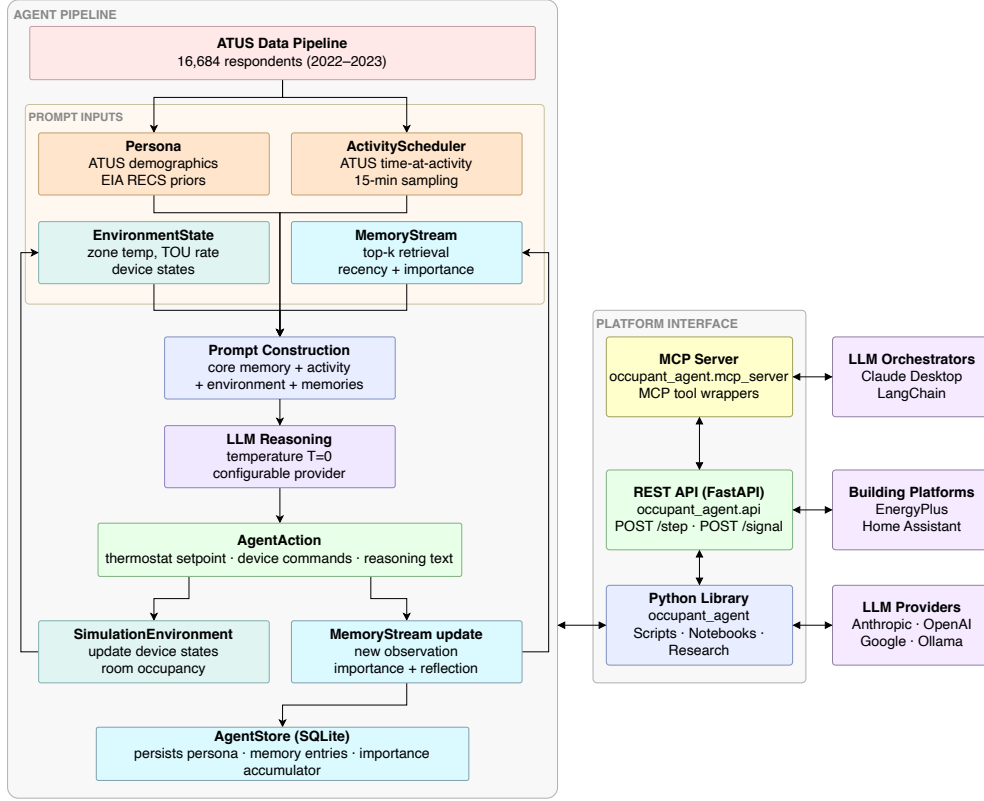


Figure 1: Architecture of BUILD OCC. *Left*: per-timestep agent pipeline showing how ATUS-grounded activity codes, retrieved memories, and environment state are assembled into an LLM prompt and how the resulting **AgentAction** updates both the simulation environment and the memory stream. *Right*: three-layer platform interface through which external building energy tools and LLM orchestration frameworks access the agent.

- **LLM Reasoning Engine** — assembles a prompt from the persona description, current ATUS activity, environment state, and retrieved memories, then queries a configurable LLM provider (Anthropic, OpenAI, Google, or local Ollama) at temperature $T=0$ for reproducibility, returning a structured action with a natural-language reasoning string.
- **SimulationEnvironment** — maintains the state of the physical building between timesteps: device on/off states, room occupancy, and thermostat setpoint. At the start of each timestep, it exposes the current state to the LLM Reasoning Engine as the environment observation; at the end, it applies the agent’s returned action and updates its internal state for the next timestep.

- **AgentStore** — persists the agent’s cognitive state — persona, all memory entries, and the importance accumulator — to SQLite via SQLAlchemy after each operation. Environment state is owned by the caller and passed fresh with each request, enabling a stateless REST API pattern that supports multi-process deployment without additional state management.

2.1.1. Platform Interface

BUILD OCC exposes the agent through three integration layers, each targeting a different class of integrator:

1. **Python library** (`occupant_agent`) — provides in-process Python access to all agent components with no server overhead, making it the recommended choice for batch simulations, Jupyter notebook experiments, and research workflows that need direct API access or custom component construction.
2. **REST API** (`occupant_agent.api`) — a stateless FastAPI [18] server exposing four HTTP operations: agent initialization, per-timestep behavioral decisions (`POST /step`), demand response signal delivery (`POST /signal`), and state query (`GET /state`). Suitable for EnergyPlus co-simulation callbacks and any building platform that can issue HTTP requests.
3. **MCP server** (`occupant_agent.mcp_server`) — wraps the REST API as tools conforming to the Model Context Protocol (MCP) [19], a standard interface that allows LLM applications to invoke external services. Any MCP-compatible host — Claude Desktop, LangChain, or Home Assistant — can drive a BUILD OCC simulation without writing additional integration code.

2.2. ATUS Behavioral Grounding

Activity schedules are derived from ATUS 2022–2023 microdata (16,684 respondents). Time-at-activity probabilities are aggregated across eight categories (sleeping, work, food preparation, laundry, television viewing, eating, exercise, and other) at 15-minute resolution, stratified by demographic group and day type (weekday / weekend). The method follows the “what is the person doing at this moment?” framing: for each diary episode $[t_{\text{start}}, t_{\text{stop}}]$, the method queries whether the episode covers the half-hour mark of each hour h , then computes a weighted probability across respondents using ATUS person-level weights.

Four demographic strata are defined (Table 2), selected to span the variation in building occupancy and demand response participation that is most relevant to residential energy research (Appendix C).

Table 2: Demographic strata

ID	Label	ATUS filter
O1	Employed single adult	Full-time employed, lives alone, age 25–44
O2	Retired couple	Age 65+, not employed
O3	Employed parent	Full-time employed, children present, age 35–54
O4	Unemployed adult	Not employed, age 25–44

An important methodological choice is the *in-progress* framing: rather than sampling from the start-time distribution (which biases toward short activities), the method evaluates which activity is in progress at the half-hour mark of each hour, reflecting the actual time spent in each activity state rather than the frequency of starts. The formal probability computation is described in Appendix A.

Work-from-home (WFH) behavior is handled separately: a per-agent WFH probability encodes the fraction of working days spent at home (drawn from ATUS 2022–2023 telework questions), and a per-day Bernoulli draw determines WFH status for each simulated day. The WFH flag is included in the agent’s per-timestep prompt so that the LLM reflects both the ATUS activity code and the work location, preventing the scheduling model from assigning “work” activities that place the agent outside the home on days it is working remotely.

Each stratum uses stratum-specific appliance ownership priors from EIA RECS [17] and an income-bracket-modulated comfort tolerance (in °C), reflecting the documented tendency of lower-income households to accept wider thermal deviations to reduce HVAC costs [20, 17]. Income brackets are currently sampled uniformly within each stratum’s RECS income range.

2.3. User-Provided Inputs

Table 3 lists all inputs required at initialization; the only per-timestep value the calling application must supply is zone temperature, since it depends on the building’s physical response to occupant actions. To lower the barrier to first use, the library ships preset helpers for a representative household device inventory and room layout. A command-line interface also ships preset helpers and a YAML-driven configuration path for users who prefer not to write Python.

Users without a live co-simulation link can supply a constant zone temperature value (suitable for behavioral experiments where thermal dynamics are not the focus) or a pre-computed CSV schedule indexed by datetime (e.g.,

Table 3: User-provided inputs to BUILD OCC. All are specified once at initialization, except zone temperature, which the calling platform supplies at each timestep.

Input	Type	Description
Stratum	String	Demographic profile: O1–O4
Seed	Integer	Controls persona attribute sampling, activity scheduling, and work-from-home day draws; the same seed produces an identical agent across runs.
LLM provider	String	Anthropic, OpenAI, Google, or Ollama
Device inventory	List	Appliance IDs, initial on/off states, rated power (W)
Room list	List	Rooms the occupant can occupy
Thermostat setpoint	Float	Initial comfort setpoint (°C)
Zone temperature	Source	Fixed value, CSV file, or live building platform
Outdoor temperature	Source	Fixed value, CSV file, EPW file, or callable
TOU rate schedule	Source	Fixed value, CSV file, or tariff function

from a prior building energy simulation run or thermostat export); a sample file covering one week of EnergyPlus output is included in the repository. Outdoor temperature is specified once at initialization; supported sources include a fixed value, a CSV schedule, a user-supplied callable, or an EnergyPlus Weather (EPW) file.

2.4. LLM Reasoning Pipeline

With all inputs initialized, at each 15-minute simulation timestep the agent executes the following five-step pipeline:

- Activity resolution:** the current ATUS code is translated to a natural-language description and an occupancy flag using a lookup table of ATUS activity codes.
- Memory retrieval:** up to five entries are selected by a retrieval score that combines recency (exponential decay, half-life 24 h) and a 1–10 importance score the LLM assigns to each entry at the time of recording — ensuring that consequential events remain retrievable even as they age (see Appendix B), following Park et al. [10].
- Prompt construction:** a system prompt encodes the persona’s stable identity — demographics, appliance ownership, and comfort tolerance.

A user prompt encodes the current environment state, ATUS activity, and the retrieved memories.

4. **LLM call:** the prompt is sent to a configurable provider (Anthropic Claude, OpenAI GPT, Google Gemini, or Ollama) at temperature $T=0$ for reproducibility. The model selects from four action types (Table 4), returning a JSON object with target, value, and a one-sentence memory note.

5. **Memory update:** the observation and its LLM-assigned importance score are appended to the stream. If the cumulative accumulator exceeds a threshold (default 100), a reflection pass is triggered in which a higher-capability model distills recent observations into three higher-order insights stored as reflection entries.

Table 4: The four action types returned by `step()` at each timestep, with their decision trigger and a representative example. Demand response is handled separately via `receive_signal()` (Section 2.5).

Action type	Trigger	Example
<code>do_nothing</code>	No environmental or activity condition warrants intervention	Agent is sleeping; no device is running
<code>adjust_thermostat</code>	Zone temperature deviates from comfort tolerance, or TOU rate context	Raises setpoint during peak window to reduce HVAC cost
<code>toggle_device</code>	Activity–appliance alignment (on) or inactivity (off)	Turns off lights when leaving for work
<code>move_room</code>	Activity context suggests transitioning to a different room	Moves to bedroom at 23:00 for sleeping

The pipeline above governs autonomous per-timestep behavior; demand response scenarios additionally require the ability to deliver external signals to the agent, which is handled by the interface described in the following subsection.

2.5. Demand Response Signal Interface

Alongside the per-timestep pipeline, BUILD OCC provides a dedicated signal delivery interface that can be invoked at any point in a simulation. Three signal types are supported (Table 5), following established behavioral intervention typologies [20, 21]. Signals are delivered by passing a signal type, a free-text message, and the current environment state to the agent, which

returns a response code (*accepted*, *rejected*, or *deferred*) alongside a natural-language reasoning string drawn from the agent’s memory context. Keeping the three types mutually exclusive enables factorial experimental designs in which signal type and demographic stratum are the primary independent variables.

Table 5: Demand response signal types supported by BUILD OCC, classified by behavioral lever and illustrated with representative example messages. All three types share the same delivery mechanism and response codes (*accepted*, *rejected*, *deferred*).

Type	Behavioral lever	Example message
A — Direct command	Regulatory compliance	“Turn off your HVAC for 30 min to support grid stability during today’s peak demand event.”
B — Educational	Economic information	“Your HVAC costs 3× more per kWh at the current peak rate — raising the setpoint 1°C saves an estimated \$0.35 today.”
C — Social norm	Peer comparison	“75% of similar households in your area have reduced HVAC use during today’s peak event.”

The extensibility mechanisms that allow all of these components to be adapted for new research contexts are described in the following subsection.

2.6. Platform Extensibility

BUILD OCC is designed for adaptation across research contexts without modifying the core library. Extensibility is organized along three independent axes:

- **Occupant population** (`BasePersona`): new demographic profiles — commercial occupants, hotel guests, non-US populations, or any custom stratum — are registered by implementing `BasePersona`.
- **Activity grounding** (`BaseScheduler`): the built-in scheduler uses ATUS microdata; any other grounding source (a Markov chain, a physics-based occupancy model, or alternative activity data) can be substituted by implementing `BaseScheduler`.
- **Memory architecture** (`BaseMemoryStream`): the built-in recency-plus-importance retrieval can be replaced with embedding-based retrieval or any other scheme by subclassing `BaseMemoryStream`.

Extensions are registered via Python’s package entry-point mechanism and discovered automatically at import time, allowing third-party packages to extend the platform without forking the repository:

Listing 1: pyproject.toml entry-point declarations for a third-party stratum and scheduler.

```

230 1 [project.entry-points."occupant_agent.strata"]
2   P5 = "my_package.personas:LowIncomeElderlyAlone"
3
14 4 [project.entry-points."occupant_agent.schedulers"]
235 5 homer = "my_package.schedulers:HomerScheduler"

```

Researchers can inject additional context into the LLM reasoning pipeline — such as predicted mean vote (PMV) comfort indices [22] or indoor air quality readings — either by extending the agent class or by passing a free-text string at each timestep; both paths are available through the REST API and MCP server. The environment state and signal schemas are fixed to ensure reproducibility, and a test utilities module provides mock implementations and conformance tests for extension authors.

3. Illustrative Examples

To illustrate agent behavior during a residential demand response event, this section traces an O1 agent (28-year-old employed male, income bracket 9, comfort band $\pm 1.1^\circ\text{C}$) through the peak electricity period of a summer weekday evening (18:00–19:45, TOU rate \$0.22/kWh). Zone temperatures come from pre-computed EnergyPlus output for the DOE PNNL residential prototype building [23] (single-family, Climate Zone 2B, variable-speed heat pump) included in the repository. The simulation loop is:

Listing 2: Running the BuildOcc simulation loop with the sample zone temperature CSV.

```

1 # Full imports available in examples/simulation_loop.py (
2   repository)
255 2 from datetime import datetime, timedelta
3
4 agent      = OccupantAgent.from_stratum("01", seed=42)
5 scheduler  = ActivityScheduler(stratum="01", seed=42)
6 zone_temp  = zone_temp_from_csv(...) # pre-computed
260   EnergyPlus zone temperatures
7
8 sim = SimulationEnvironment(
9     initial_devices=persona_devices(agent.persona),
10    initial_rooms=[RoomState("living_room", True)],
265 1    thermostat_setpoint=22.0,

```

```

12     outdoor_temp_fn=summer_day_temp,
13     tou_rate_fn=peak_tou_rate,
14 )
15
270 6 base = datetime(2025, 8, 10, 18, 0)
17 for i in range(8):
18     ts      = base + timedelta(minutes=15 * i)
19     env      = sim.observe(ts, zone_temp(ts))
20     action   = agent.step(env, atus_code=scheduler.sample(
275         ts))
21     sim.apply(action, ts)
22     print(ts.strftime("%H:%M"), action.action_type,
           action.reasoning)

```

280 At 18:00 (Step 1), the CSV-sourced zone temperature is 23.9°C and the thermostat is set to 22°C. With outdoor temperatures exceeding 38°C, the HVAC is running continuously to hold the setpoint. The agent is engaged in computer leisure and proactively raises the thermostat setpoint to 24°C to reduce peak-rate HVAC run time, reasoning:

285 *“The indoor temp is 23.9°C and the thermostat is set to 22°C — the HVAC is running hard to maintain the setpoint with 39°C outside during peak rate hours (\$0.22/kWh). I’m comfortable at 23.9°C, well within my 1.1°C tolerance, so raising the setpoint to 24°C will meaningfully cut cooling costs during this expensive period.”*

This response demonstrates that the persona’s income bracket and comfort tolerance surface in actionable decision-making without explicit rule encoding.

295 At 18:30 (Step 3), a Type B educational signal is delivered: *“Your HVAC is running during peak hours (4–9pm) at \$0.22/kWh—three times the off-peak rate. Raising the setpoint by 1°C saves ~\$0.35 today.”* The agent accepts the signal but references its own prior action stored in memory:

300 *“I already raised the thermostat to 24°C earlier today and it’s working—I’m at 23.9°C and comfortable within my tolerance. The math is clear (\$0.35 savings) and the effort is zero since I’ve already taken this action.”*

This cross-timestep memory coherence—where the agent recalls and references its own prior decision—illustrates the emergent behavioral consistency that distinguishes LLM-grounded agents from rule-based or lookup-table

baselines. A fixed schedule model could encode a thermostat raise at 18:00, but it cannot encode the nuance that the agent chose that action for cost reasons and remembers having already acted when a subsequent signal arrives.

Over the remainder of the evening, the agent maintains the raised setpoint without further intervention. Importance ratings for routine observations are low (2–3 out of 10), correctly prioritizing memory budget toward the salient cost-saving event. By contrast, an O4 agent (unemployed adult) carries a fuller day-long memory of peak-rate exposure and, in representative runs, accepts the same Type B signal citing consistency with prior device-toggling decisions rather than comfort margin—a qualitatively different rationale that reflects demographic grounding rather than explicit rule differences.

Figure 2 illustrates the behavioral fidelity of the activity scheduler for stratum O1 (employed single adult) on weekdays. The ATUS-grounded simulation (panel a) closely tracks the faint reference lines, while the fixed rule-based baseline (panel b) diverges markedly — most notably collapsing all daytime activity into sharp step functions and failing to reproduce the gradual evening wind-down.

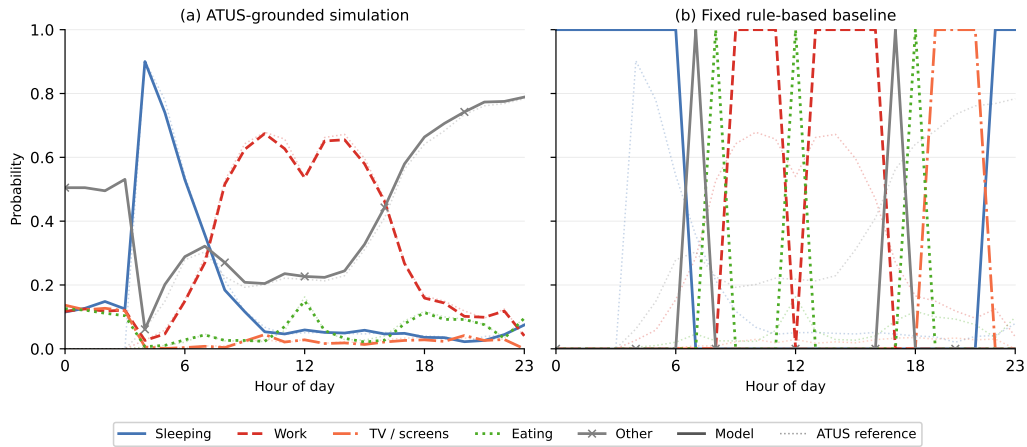


Figure 2: Hourly activity probability distributions for stratum O1 (employed single adult, weekday) across five activity categories. (a) ATUS-grounded simulation (180 simulated days): bold lines closely track the reference, confirming scheduler fidelity. (b) Fixed rule-based baseline: bold lines diverge sharply, with the baseline collapsing smooth activity transitions into unrealistic step functions and misassigning the O1 late-evening activity peak.

4. Validation

Validation is organized in two tiers targeting different components of the platform. Tier 1 validates the *ActivityScheduler*: it tests whether ATUS-grounded activity sampling produces category distributions that match empirical ATUS reference data, relative to a deterministic rule-based baseline. Tier 2 validates the *LLM agent decisions*: it tests whether the demographic priors encoded in each stratum’s persona propagate into distinguishable behavioral patterns under identical environment conditions. Both tiers are necessary because a well-grounded scheduler and a well-calibrated reasoning agent are independent failure modes — one can be correct while the other is not.

4.1. Tier 1: Activity Schedule Fidelity

The ATUS-grounded activity scheduler is validated against a deterministic rule-based baseline using KL divergence between simulated activity distributions and ATUS empirical distributions. For each stratum, 180 days of 96 timesteps each (17,280 observations per stratum) are simulated, the empirical category distribution at each hour is computed, and compared to the ATUS reference using:

$$D_{\text{KL}}(P\|Q) = \sum_c P(c) \log \frac{P(c)}{Q(c) + \varepsilon} \quad (1)$$

where P is the ATUS reference distribution, Q is the simulated distribution, and $\varepsilon = 10^{-9}$ is a smoothing constant. Table 6 and Figure 3 report the mean per-hour KL divergence averaged across all 24 hours and both day types. Figure 2 (preceding section) shows the corresponding hourly distributions for O1, illustrating how closely the ATUS-grounded scheduler tracks the empirical reference compared to the fixed baseline.

Table 6: Mean per-hour KL divergence (nats): ATUS-grounded vs. rule-based scheduler. Computed over 180 simulated days, 96 timesteps each, weekday/weekend split. Lower is better.

Stratum	ATUS scheduler	Fixed baseline
O1 (Employed single adult)	0.0181	10.11
O2 (Retired couple)	0.0253	11.48
O3 (Employed parent)	0.0184	9.92
O4 (Unemployed adult)	0.0092	11.27

The fixed baseline assigns deterministic activities by hour-of-day rules (e.g., sleeping 23:00–06:59, work 09:00–16:59) without demographic conditioning,

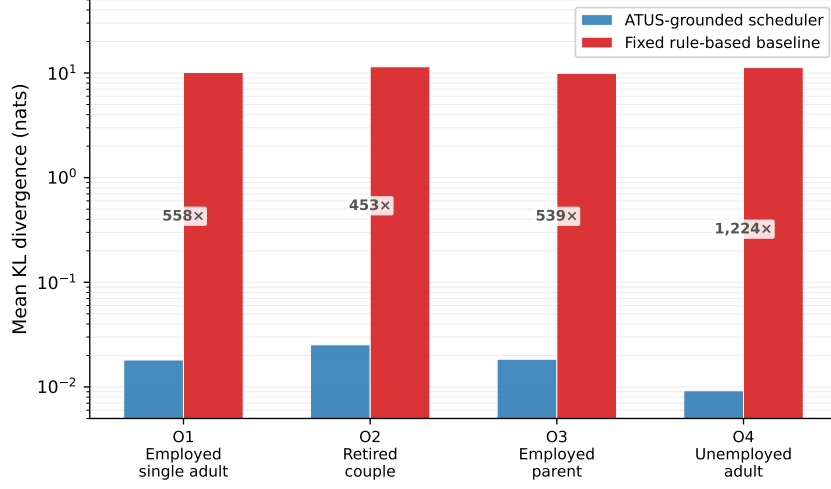


Figure 3: Mean per-hour KL divergence (nats, log scale) between simulated and ATUS empirical activity distributions over 180 simulated days. Annotated ratios indicate the fold-reduction achieved by the ATUS-grounded scheduler relative to the fixed rule-based baseline.

which produces large divergence against the ATUS reference—especially for the retired (O2) and unemployed (O4) strata, whose schedules differ most from the employed-adult assumption. The 1,224-fold reduction in KL divergence for O4 reflects the extreme mismatch between a generic nine-to-five work rule and the ATUS-observed daytime activity patterns of unemployed adults.

4.2. Tier 2: Persona-Driven Behavioral Differentiation

Tier 2 tests whether the demographic priors encoded in each stratum’s persona produce distinguishable behavioral patterns under identical environment conditions, confirming that persona variation — not environment variation — is the primary driver of the observed differences. Each of O1–O4 was run for three independent seeds over one weekday (96 timesteps at 15-minute resolution) under identical environment conditions—the same EnergyPlus zone temperature, device configuration (HVAC, three lighting zones, TV, washer, dishwasher), and time-of-use (TOU) tariff (\$0.22/kWh peak, \$0.08/kWh off-peak)—so that behavioral differences are attributable solely to persona variation.

Table 7 reports the resulting action-type distributions and mean thermostat setpoints during the TOU peak window (4–9 pm). O2 (retired couple) exhibits the highest overall activity level—lowest inactivity rate (67.4%), highest device-toggling rate (8.0%), highest room-movement rate (21.5%), and

highest mean peak setpoint (24.3°C)—consistent with full-day home occupancy and active use of three configurable lighting zones as the agent transitions between kitchen, dining room, and living room throughout the day. O4 (unemployed adult; ATUS $n=107$, Appendix C) matches O2’s toggling rate (8.0%) but records the lowest thermostat-adjustment rate (1.4%) and the lowest peak setpoint (23.1°C), suggesting cost-sensitive passive thermostat management despite active device engagement. O3 (employed parent) shows the lowest device-toggling rate (3.5%): ATUS child-care codes (physical care, play, and supervision activities) dominate O3’s peak-hour schedule, reducing behavioral attention available for appliance and thermostat control. O1 (employed single adult) sustains a moderate profile across all action types, with an inactivity rate of 76.0% and the second-highest peak setpoint (23.5°C). Figure 4 visualizes these distributions.

Table 7: Action-type distribution (%) per stratum: mean over three seeds, one weekday each (96 steps). Peak SP is the mean thermostat setpoint during TOU peak hours (4–9 pm, \$0.22/kWh).

Stratum	Inactive	Thermostat	Toggle	Move	Peak SP (°C)
O1 (Employed single)	76.0	2.8	4.2	17.0	23.5
O2 (Retired couple)	67.4	3.1	8.0	21.5	24.3
O3 (Employed parent)	76.4	3.1	3.5	17.0	23.6
O4 (Unemployed)	72.9	1.4	8.0	17.7	23.1

To assess demand response participation, three standardized signals were delivered to each stratum (five seeds each) after a 2.5-hour context period at peak-rate conditions (10 simulated timesteps beginning at 16:00, giving each agent an experiential baseline in memory before signal delivery): Type A (direct command: “Turn off your HVAC for 30 minutes”), Type B (educational price signal: cost comparison at three times the off-peak rate), and Type C (social norm: “75% of similar households have reduced HVAC use”). Table 8 reports accept rates per stratum and signal type. Direct commands (Type A) achieved the highest acceptance for O2 (retired couple, 40%), with O1 (employed single, 20%) and O4 (unemployed adult, 20%) showing moderate compliance; educational price signals (Type B) matched this rate for O2 and O4 (40% each), suggesting that cost-aware strata respond to both authority and economic framing. O3 (employed parent) rejected all signal types across all five seeds: signal delivery at 18:30 coincides with child-care activity codes in the ATUS schedule, making the 0% acceptance rate an expected consequence of schedule grounding rather than a general insensitivity to demand response. Social norm signals (Type C) achieved zero acceptance across all strata, consistent with field evidence on the limited efficacy of

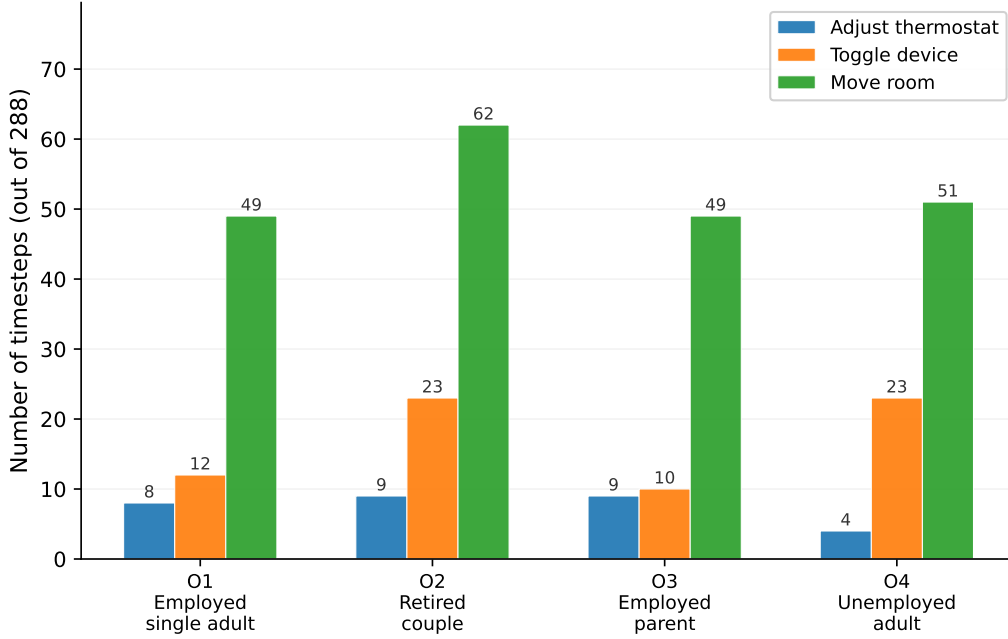


Figure 4: Active action counts per stratum across 288 timesteps (3 seeds $\times$ 96 steps, identical environment conditions). Do-nothing accounts for 67–76% of all timesteps across strata and is excluded to isolate cross-stratum differences in active decision-making. Bar labels show absolute counts.

peer-comparison messaging [20, 21].

Table 8: Signal compliance rates (accept %) per stratum and signal type over five random seeds. Type A: direct command; Type B: educational price signal; Type C: social norm nudge.

Stratum	Type A	Type B	Type C
O1 (Employed single)	20	0	0
O2 (Retired couple)	40	40	0
O3 (Employed parent)	0	0	0
O4 (Unemployed)	20	40	0

Figure 5 compares thermostat setpoint trajectories of O1 and O3 over the simulated day. O1 agents operate at setpoints averaging 0.5°C higher than O3 during peak hours; the single adult’s narrow comfort band (1.1°C) and absence of family obligations enable individual thermostat optimization, while O3’s schedule is dominated by child-care activities during the peak window.

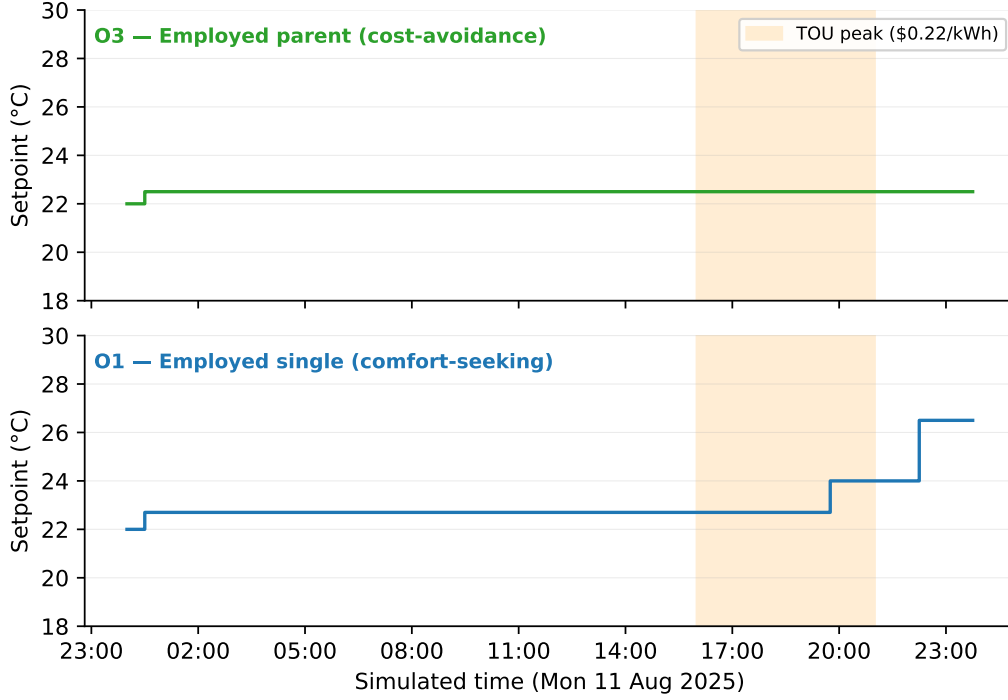


Figure 5: Thermostat setpoint trajectories for O3 (employed parent, family-constrained, upper panel) and O1 (employed single adult, comfort-seeking, lower panel). Shaded region: TOU peak period (4–9 pm). O1 agents sustain setpoints averaging 0.5°C higher than O3 during peak hours, reflecting individual cost-avoidance unconstrained by family obligations.

These results confirm that the ATUS demographic priors propagate into qualitatively distinct LLM-driven behavioral patterns: employment status shapes thermostat adjustment frequency, household composition modulates signal compliance, and full-day home occupancy drives device-toggling behavior. Two design constraints bound the interpretation. First, each stratum is run for only one weekday per seed (three seeds for action distributions, five for signals), which is sufficient to show directional patterns but not to establish statistical significance; multi-day designs are feasible but were constrained by API cost. Second, Tier 2 uses a single environment configuration (fixed weather, tariff structure, and household composition), which limits generalizability across different climate zones and occupancy contexts. The behavioral differences across strata also reflect a structural property of the platform: both the ATUS-grounded activity schedule and the LLM persona reasoning are conditioned on the same demographic stratum, a coupling that ensures in-

ternal consistency between the scheduling and decision-making components.

5. Impact

BUILD OCC enables four lines of research that were previously impractical with fixed-schedule occupant models:

Survey-grounded behavioral agents. BUILD OCC makes population-calibrated occupant agents a shared community resource rather than a per-project infrastructure investment. Research groups can now introduce realistic, demographically differentiated occupant behavior into any building energy study without constructing a grounding pipeline from scratch. The ATUS-derived activity distributions and RECS-conditioned personas provide an empirically validated behavioral baseline that any researcher can adopt, extend, or cite as a reference condition — shifting what was previously a specialized capability into a reusable foundation available to the entire building energy community.

Demographic behavioral experiments. BUILD OCC’s combination of ATUS-grounded demographic strata, persona income brackets, and configurable intervention signals enables controlled factorial studies of occupant behavior. Unlike fixed-schedule models, each agent reasons from its demographic persona and accumulated memory at every timestep, producing adaptive responses that reflect personal history and current context rather than predetermined patterns. Because both the activity schedule and the agent’s reasoning are conditioned on the same demographic persona, researchers can hold environment and intervention constant while varying stratum — or vary the intervention while holding stratum constant — isolating the contribution of socioeconomic and life-stage factors to any behavioral outcome. The Tier 2 signal experiment (Table 8) illustrates this: Type B educational signals achieved equal or higher acceptance than direct commands for cost-sensitive strata (O2 and O4: 40% each), while O1 (employed single adult) responded only to Type A direct commands (20%) and O3 (employed parent) rejected all signal types — compliance gradients that fixed-schedule models cannot reproduce. The signal types correspond to established intervention typologies in behavioral energy research [20, 21], making results directly comparable to field studies.

Platform-neutral integration. The MCP server layer means that any tool with MCP support—including Claude Desktop, LangChain, energy scheduling systems [24], and custom orchestration frameworks—can drive a building simulation with realistic occupant behavior without writing integration code. The REST API provides an equivalent path for EnergyPlus co-simulation and Home Assistant integrations [5, 8].

Community extensibility. The plugin registry and abstract base classes lower the barrier for researchers who want to add custom demographic strata, alternative behavioral data sources or activity grounding pipelines, or novel memory architectures. The conformance test suite enables rapid validation of extensions before publication.

6. Conclusions

BUILD OCC delivers a survey-grounded, three-layer occupant agent platform that the building energy community can use, extend, and build upon without reconstructing the behavioral layer for each study. Two validation tiers confirm that ATUS grounding produces empirically calibrated activity distributions and that demographic priors propagate into meaningfully differentiated agent reasoning under environmental and intervention signals.

The platform’s deeper contribution is a reframing: occupant behavior shifts from an uncontrolled source of uncertainty to a structured, demographically parameterizable experimental variable. The ATUS-derived strata, frozen schema, and plugin registry together make that variable reproducible and extendable across independent research groups — a property that has been absent from the occupant behavior modeling literature.

Current limitations include a single-action constraint per timestep and ATUS-grounded strata covering US demographic profiles only. Subsequent phases extend the platform with multi-agent household simulation, non-US occupancy data, and external behavioral benchmarks including ecobee Smart Thermostat field data [25] matched to the four ATUS strata.

CRedit Author Statement

Wooyoung Jung: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The author declares that there are no competing interests.

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A. ATUS Activity Probability Computation

This appendix provides the formal description of the activity probability computation summarized in Section 2.

Episode-overlap framing. ATUS diaries record each respondent’s day as a sequence of episodes, each with a start time t_s (minutes since midnight), stop time t_e , and a six-digit activity code in ATUS. For a 15-minute simulation slot centered at time τ , the scheduler identifies which activity episode was in progress at the midpoint $\tau + 7.5$ min:

$$c(r, \tau) = \text{activity code of respondent } r \text{ such that } t_s \leq \tau + 7.5 \text{ min} < t_e \quad (\text{A.1})$$

where r indexes individual diary respondents. This in-progress query avoids the start-time bias that would overrepresent short activities (e.g., a 5-minute meal counts only if the midpoint falls within it).

Category probability. Eight activity categories are defined (Table A.9). The probability of category k at hour h , for demographic stratum s and day type $d \in \{\text{weekday, weekend}\}$, is:

Table A.9: The eight broad activity categories used in the two-step ATUS sampling procedure, with their ATUS code filter, the number of Tier-3 codes mapped to each category, and representative subcategories.

Category	ATUS code filter	Codes	Representative subcategories
Sleeping	prefix 0101	3	Sleeping; Sleeplessness
Work	prefix 05	8	Work (main job); Work (other job); Socializing at job
Food preparation	prefix 0201	4	Food/drink preparation; Food presentation; Kitchen clean-up
Laundry	code 020202 exactly	1	Laundry
Television	code 120301 exactly	1	Watching TV
Eating	prefix 1101	2	Eating and drinking
Exercise	prefix 13	29	Aerobics; Basketball; Biking; Swimming; Weightlifting
Other	residual	172	Grooming; Reading; Computer use; Social communication; Travel

$$P(k \mid s, d, h) = \frac{\sum_{r \in R(s, d)} w_r \cdot \mathbf{1}[c(r, h:30) \in k]}{\sum_{r \in R(s, d)} w_r} \quad (\text{A.2})$$

where $R(s, d)$ is the set of diary respondents matching stratum s on day type d ; w_r is the ATUS person-level survey weight (provided by BLS to correct for sampling design); $\mathbf{1}[\cdot]$ is the indicator function that equals 1 when respondent r was engaged in a category- k activity at the half-hour mark $h:30$ of hour h ; and the denominator normalizes the weighted counts to probabilities. Evaluating Equation (A.2) for all $h \in \{0, \dots, 23\}$ and all 8 categories yields a 24×8 weighted probability table per stratum per day type, which is bundled with the installed package.

Code-level sampling. Once a category k is sampled from $P(k \mid s, d, h)$, a specific activity code is drawn from within that category proportional to the total diary minutes logged across all stratum respondents for each code. This gives common activities realistic relative frequencies (e.g., sleeping in a bed is drawn far more often than napping on a sofa, even though both map to the “sleeping” category).

Work-from-home probability. Finally, the WFH probability p_{WFH} for each stratum is estimated from the ATUS `TEWHERE` variable — the “where were you when you did this activity?” field — restricted to employed respondents who worked on their diary day. p_{WFH} equals the survey-weighted fraction of those respondents who reported working at home for any work episode, computed separately for each stratum. A per-day Bernoulli draw from p_{WFH} then determines whether the simulated agent works from home on a given day.

B. Memory Retrieval and Reflection Mechanics

This appendix provides the mathematical detail for the memory retrieval and reflection logic implemented in `MemoryStream`, following the generative agent architecture of Park et al. [10].

Retrieval scoring. At each simulation timestep t , every stored memory entry m receives a combined retrieval score $s(m, t) \in [0, 1]$ that balances how recent the entry is against how important it was judged to be when written:

$$s(m, t) = 0.5 \tilde{r}(m, t) + 0.5 \tilde{i}(m) \quad (\text{B.1})$$

where $\tilde{r}(m, t)$ is the normalized recency score and $\tilde{i}(m)$ is the normalized importance score, both in $[0, 1]$ and weighted equally.

The recency component decays exponentially with a 24-hour half-life:

$$\tilde{r}(m, t) = 2^{-(t-t_m)/24\text{h}} \quad (\text{B.2})$$

where t_m is the simulation time (in hours) at which entry m was written, and $(t - t_m)$ is the elapsed time in hours since writing. The score equals 1.0 at the moment of creation and halves every 24 simulation hours, so memories from the previous day retain roughly half their recency weight while week-old entries contribute near zero.

The importance component normalizes the integer score $i(m)$ assigned by the LLM at write time:

$$\tilde{i}(m) = \frac{i(m)}{10} \quad (\text{B.3})$$

where $i(m) \in \{1, \dots, 10\}$ is an integer on a scale from 1 (routine, nothing unusual) to 10 (rare, extreme situation), self-assigned by the agent’s LLM in the same JSON response that produces the behavioral action. The top- k entries ranked by $s(m, t)$ are injected into the next prompt ($k = 5$ by default).

Reflection trigger. An importance accumulator A tracks the cumulative importance of all entries written since the last reflection. After each entry m is stored, $A \leftarrow A + i(m)$. When $A \geq \theta$ (default $\theta = 100$), reflection is triggered: a dedicated LLM call receives the 30 most recent entries by simulation time and returns exactly three synthesized behavioral insights (e.g., “this occupant pre-cools before peak hours to avoid peak tariffs”). Each insight is written back into the stream as a new entry with importance 9.0, and A resets to zero. At average importance 7–10 (high-salience events), reflection fires roughly every 10–14 timesteps (≈ 2.5 – 3.5 simulation hours); at average importance 2–3 (routine observations), every 33–50 timesteps (≈ 8 – 12 simulation hours).

C. Demographic Strata Selection Rationale

The four strata were selected to maximize variation in the two behavioral dimensions most consequential for residential demand response research: *peak-hour occupancy schedule* and *household composition*. The selection is grounded in the ATUS 2022–2023 microdata that drives the platform itself.

Table C.10 summarises the occupancy metrics and sample sizes derived from ATUS 2022–2023 ($n = 16,684$ respondents). O1 and O3 are absent from the home for the majority of peak grid hours (09:00–17:00), while O2 and O4 are home throughout the day — a 14-fold difference in daytime building presence.

Within each occupancy cluster, household composition is varied to capture differences in thermostat authority and social dynamics: single-person (O1, O4), two-adult couple (O2), and family with children (O3) [20, 21]. O4 has the smallest sample ($n = 107$); its 15-minute probability tables are the noisiest of the four strata. Additional strata (multi-adult non-family, student dormitories) would draw from the remaining 10,643 “other” respondents but would fragment already thin subsamples without a clearly defined DR participation profile.

Table C.10: ATUS 2022–2023 occupancy metrics and sample sizes by stratum. % working 09–17 is the average weighted probability of being engaged in work activity during peak hours on weekdays (Appendix A). Work location columns are derived from the ATUS TEWHERE variable restricted to employed respondents on diary days with a work episode.

ID	Stratum	% working 09–17 (wkdy)	% at workplace	% at home	n
O1	Employed single adult	59.8	70.2	24.3	1,393
O2	Retired couple	0.4	8.1	50.1	2,351
O3	Employed parent	61.4	68.9	25.7	2,190
O4	Unemployed adult	7.9	0.0	98.4	107

Current executable software version

Table 11: Software metadata (mandatory)

Nr.	Software metadata description	Please fill in this column
S1	Current software version	v1.0.0
S2	Permanent link to executables of this version	https://pypi.org/project/buildocc/
S3	Legal software license	Apache License 2.0
S4	Computing platforms / OS	Linux, macOS, Windows (Python 3.11+)
S5	Installation requirements & dependencies	<code>pip install buildocc</code> ; LLM provider API key required
S6	Link to user manual	https://github.com/humanbuildingsynergy/Build0cc/blob/main/README.md
S7	Support email for questions	wooyoungjung@arizona.edu

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Declaration of Interest Statement

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The author is an Editorial Board Member/Editor-in-Chief/Associate Editor/Guest Editor for this journal and was not involved in the editorial review or the decision to publish this article.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:



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07/04/2026

Dear Editor:

I am pleased to submit the manuscript "BuildOcc: A Large Language Model Occupant Agent Platform for Building Energy Research" for consideration in SoftwareX.

BuildOcc is an open-source Python platform that grounds large language model (LLM) agents in the American Time Use Survey (ATUS) — a nationally representative diary dataset covering 16,684 respondents — to simulate building occupant behavior for energy research. Existing occupant behavior models rely on fixed schedules or simple stochastic processes that cannot reason about a person's history, current context, or the type of energy signal being delivered. Prior LLM-agent proposals in this space have used hand-crafted profiles or small convenience samples and have not released a general-purpose, survey-grounded platform for reuse. BuildOcc fills this gap. Each simulated occupant maintains a persona derived from ATUS population statistics, a timestep-level memory stream with periodic reflection, and an activity scheduler that samples empirically from ATUS time-at-activity distributions. The platform exposes a three-layer interface (Python library, REST API, and Model Context Protocol server) so that any building energy tool can integrate behavioral intelligence without bespoke coupling code. A plugin registry allows the community to add new demographic strata, custom schedulers, and alternative memory backends as separate installable packages. BuildOcc is an original, openly released software contribution (Apache License 2.0, v1.0.0 at github.com/humanbuildingsynergy/BuildOcc) that addresses a concrete gap in the building energy simulation community:

We believe that this article strongly contributes to the building energy research community. Also, it fits the scope and aim of SoftwareX and complies with its requirements. This manuscript has not been published and is not under consideration for publication elsewhere.

Thank you very much for your consideration.

Sincerely,

A handwritten signature in black ink, appearing to read 'WJung'.

Wooyoung Jung, Ph.D.

Assistant Professor, The University of Arizona

